

Predicting Restaurant Failure: An NLP-Driven Risk Assessment from Yelp Reviews

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Introduction

In modern hospitality, the online reputation of restaurants is often the main determinant of business longevity and success. The Yelp Open Dataset provides a large source of customer sentiment, offering an excellent opportunity to analyze the feedback provided by the customers on a deeper level. This project explores whether the linguistic patterns and sentiments hidden within the customer reviews can identify the signs of a restaurant's potential closure.

Related Work

Different work and research papers were explored to determine which models are best suited for the given problem. One work, done by Prabhat and Khullar (2017) used Naive Bayes and Logistic Regression for sentiment classification. Naive Bayes (NB) is a classification algorithm used in machine learning that uses probability to predict the category of a data point (GeeksforGeeks, 2026), while Logistic Regression is a supervised machine learning classification algorithm used in data science that predicts a discrete or categorical outcome (Lee, 2025). In another work done by Pranckevičius and Marcinkevičius (2017), comparisons between NB, Random Forest, Support Vector Machines, and Logistic Regression were explored. Support Vector Machines (SVM) is a supervised machine learning algorithm that classifies data by finding an optimal line that maximizes the distance between each of these classes in an N-dimensional space (Kavlakoglu, 2023). Kavalakoglu (2021) also provides the definition of Random Forest, which is another machine learning algorithm that takes the output of multiple decision trees and combines them together to create a single result. Irfan et al. (2025) integrated a multiclassifier sentiment analysis framework that integrated six machine learning models, including Logistic Regression, with TF-IDF features to classify 50,000 movie reviews. Additionally, in another study by Angadi and Reddy (2019) demonstrated that by using a random forest classifier with an optimal feature selection, the accuracy of a multimodel sentiment analysis is improved compared to other existing systems.

Project Problem/Motivation Statement

The project uses a Natural Language Processing (NLP) approach to analyze reviews and look for patterns to be used to predict how likely a restaurant is to fail and close. NLP is a technology that

allows computers to understand, comprehend, and manipulate human language. The aim is to create a predictive model that will use NLP to analyze online reviews gathered from the Yelp Open Dataset and identify early warning signs of a restaurant's potential closure. The model employs different NLP techniques such as TF-IDF and sentiment analysis. The project could be beneficial for restaurant owners and investors, allowing them to have insight ahead of time regarding the success of the restaurant. This opens space for potential changes, improvements, or preparations that can lead to avoiding or preventing a restaurant failure.

Big Data Analytics Project Lifecycle Stages

a. Understand requirements

The data was downloaded as five JSON files. Using the documentation provided for it, each JSON was inspected. The overall dataset has about 6,990,280 reviews and 150,346 businesses. The review data JSON contains full review text data, including the user_id that wrote the review and the business_id the review is written for. User.json contains user data, including the user's friend mapping and all the metadata associated with the user. Checkins.json shows the checkins on a business. Tip.json shows tips written by a user on a business (tips are shorter than reviews and tend to convey quick suggestions). After inspection, the data that is needed for our model is conained in reviews.json, business.json, tips.json as each contains either the review data or the metadata about the restaurant, which are necessary for the model. Most significantly, the business-related file contains an attribute is_open which indicates if the restaurant is still operating or not. This will be the target feature.

b. Acquire data

The data was acquired from the following website: https://business.yelp.com/data/resources/opewebsiteet/. It is open source and easily downloadable. All five JSON files are downloaded at once, along with their documentation. However, since not all data is needed, the outlined three JSON files were combined into one by joining according to the business ID. Then, columns that were not needed were dropped. For instance, 'cool', 'funny', and 'useful' are the reactions to a specific review and are deemed unnecessary due to high scarcity. Features like 'latitude', 'longitude', 'postal_code', 'attributes', and 'hours' were deemed irrelevant to the goal of the model, as the focus is on the impact of reviews on restaurant success rather than

location or whether parking is available (if relevant, they will be mentioned in the reviews anyway). The 'categories' feature was used to filter for just the restaurant data in the dataset before dropping it, since the Yelp dataset includes other businesses, such as retail. The final dataset included 190 590 rows and 14 columns.

c. Preprocess data

Firstly, on a global level, null and duplicate values were dropped. There were no missing values, but there were about 500 duplicate values, which were dropped from the final dataset, decreasing its size slightly. Then, three cleaning strategies were applied. A pipeline was created for TF-IDF and for semantic analysis, as slightly different cleaning techniques are used for each. Both convert the text to lowercase and collapse extra spaces, however TF-IDF is more aggressive as it only keeps letters, while for sentiment analysis, numbers, '!', and '?' were kept due to possible sentiment value. For both URLs, emojis and emails were removed. For other, non-review columns, the function converts the city, state, address, and name columns to lowercase and trims whitespace; the address and name fields undergo further cleaning via regex to collapse multiple spaces and remove non-alphabetic characters from names. Additionally, it casts the date column to a standard date format and extracts the most recent check-in date from a potential list of timestamps. The original, unrefined columns are dropped to keep the schema clean. Additionally, to ensure optimal model performance, scaling methods were implemented using StandardScaler that centers the features and normalizes their variance. While for Random Forest scaling is not crucial, all models used scaled data to ensure consistency.

d. Exploratory Data Analysis (EDA)

For this stage, multiple graphs and charts were created, as well as tables from the joined datasets. Figure 1 displays the top 10 and bottom 10 restaurants based on the number of reviews that were left on the Yelp website.

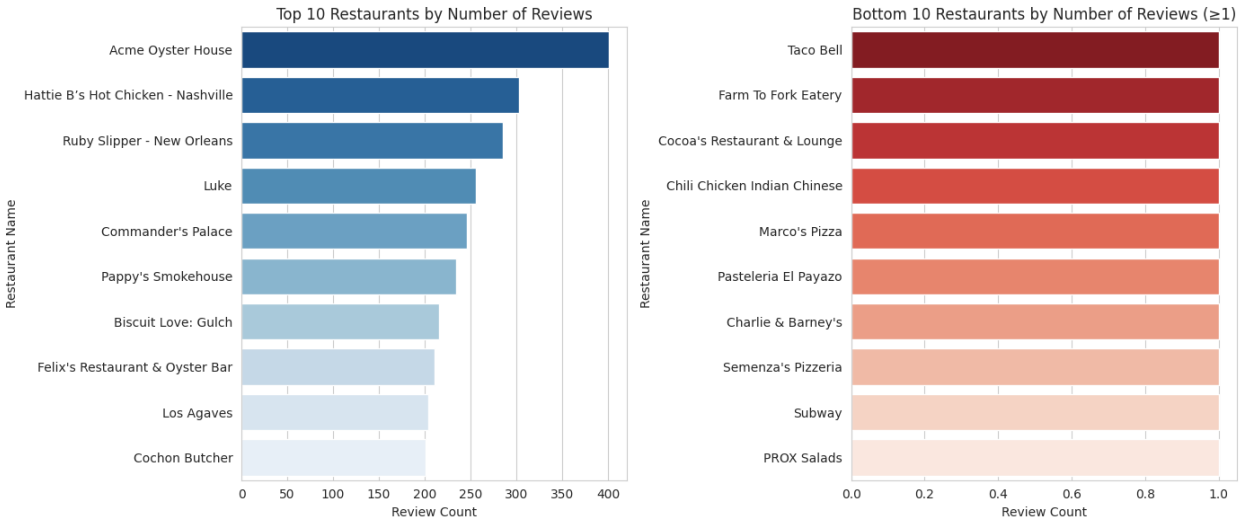


Fig. 1., Top 10 Restaurants by Number of Reviews (left) and Bottom 10 Restaurants by Number of Reviews (right)

The results show that the restaurant with the most reviews left is the "Acme Oyster House" with about 300 reviews. And the restaurant with the fewest reviews according to the plot is "Taco Bell". It is important to note that the plot on the right in Figure 1 shows that every restaurant in the dataset has at least one review. The second plot seen in Figure 2 groups businesses by the state they are in (exclusive to the United States) and then creates a chart of the top 10 states with the most businesses/restaurants. Based on the result, the state of Pennsylvania has the most restaurants with over 6000 present.

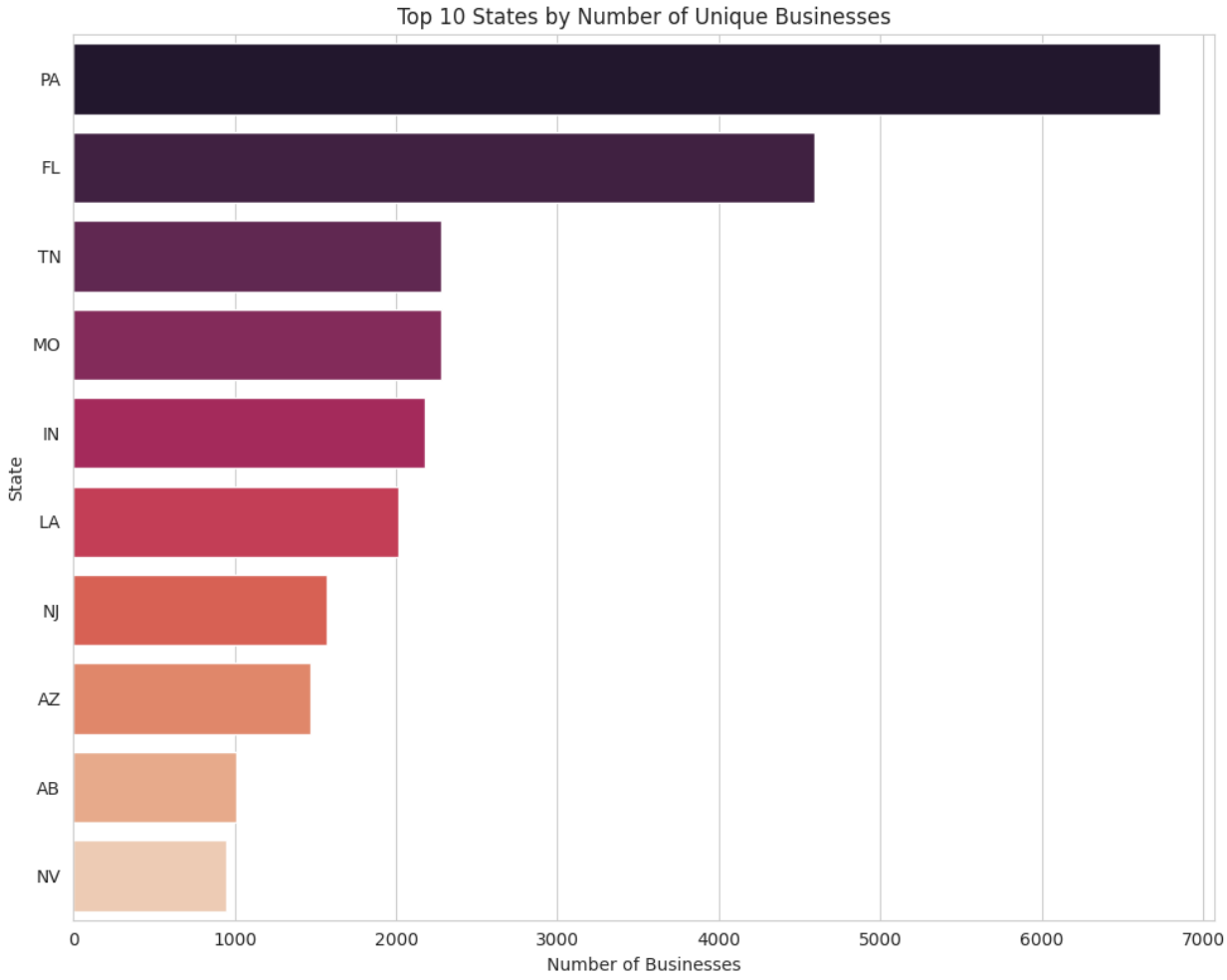


Fig.2., Top 10 States by Number of Unique Businesses

In addition to the previous plot, another one was created to display the number of restaurants in different cities within the state of Pennsylvania (Figure 3). Considering that there are restaurants located all over the state, the plot was limited to show only the top 20 for readability.

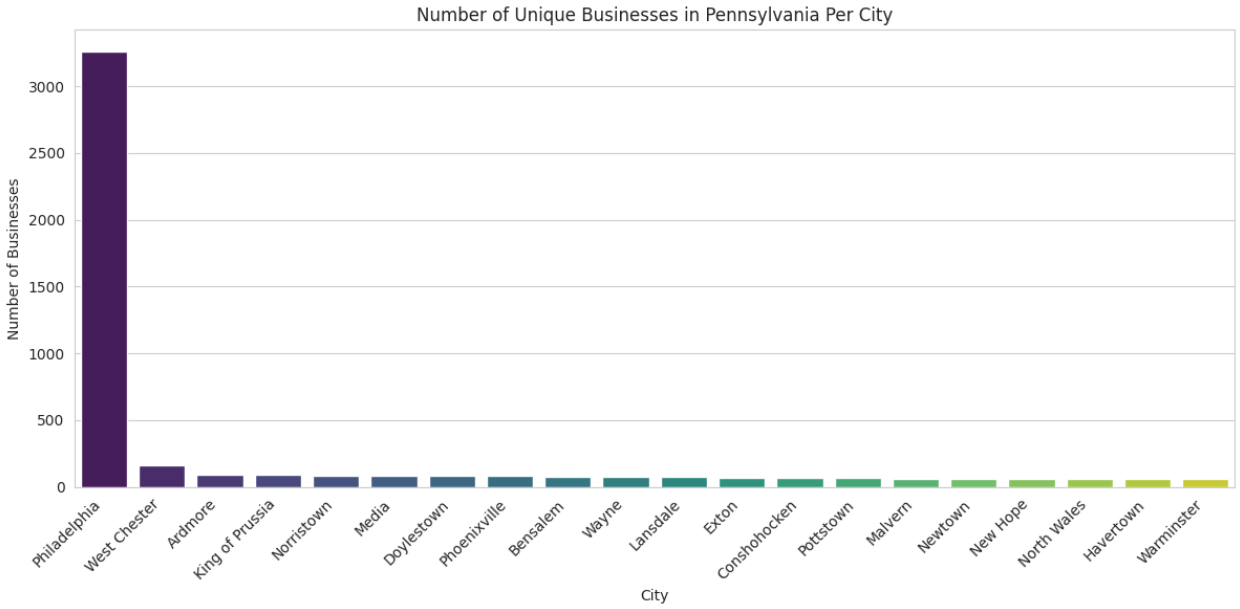


Fig.3., Number of Unique Businesses in Pennsylvania Per City

According to these results, most of the restaurants are located in Philadelphia, with over 3000 restaurants registered. There is also a significant difference compared to the other cities, as the next city with the most restaurants (West Chester), has less than 500 restaurants present in the dataset. Another visualization was done that relates directly to the star ratings of restaurants (Figure 4). These ratings use a scale of 0-5. The ratings were split into 5 groups (0-1.0, 1.5-2.0, 2.5-3.0, 3.5-4.0, 4.5-5.0). Each restaurant is then compared by rating and classified into one of these groups, to see the most common rating given to these restaurants, percentage-wise.

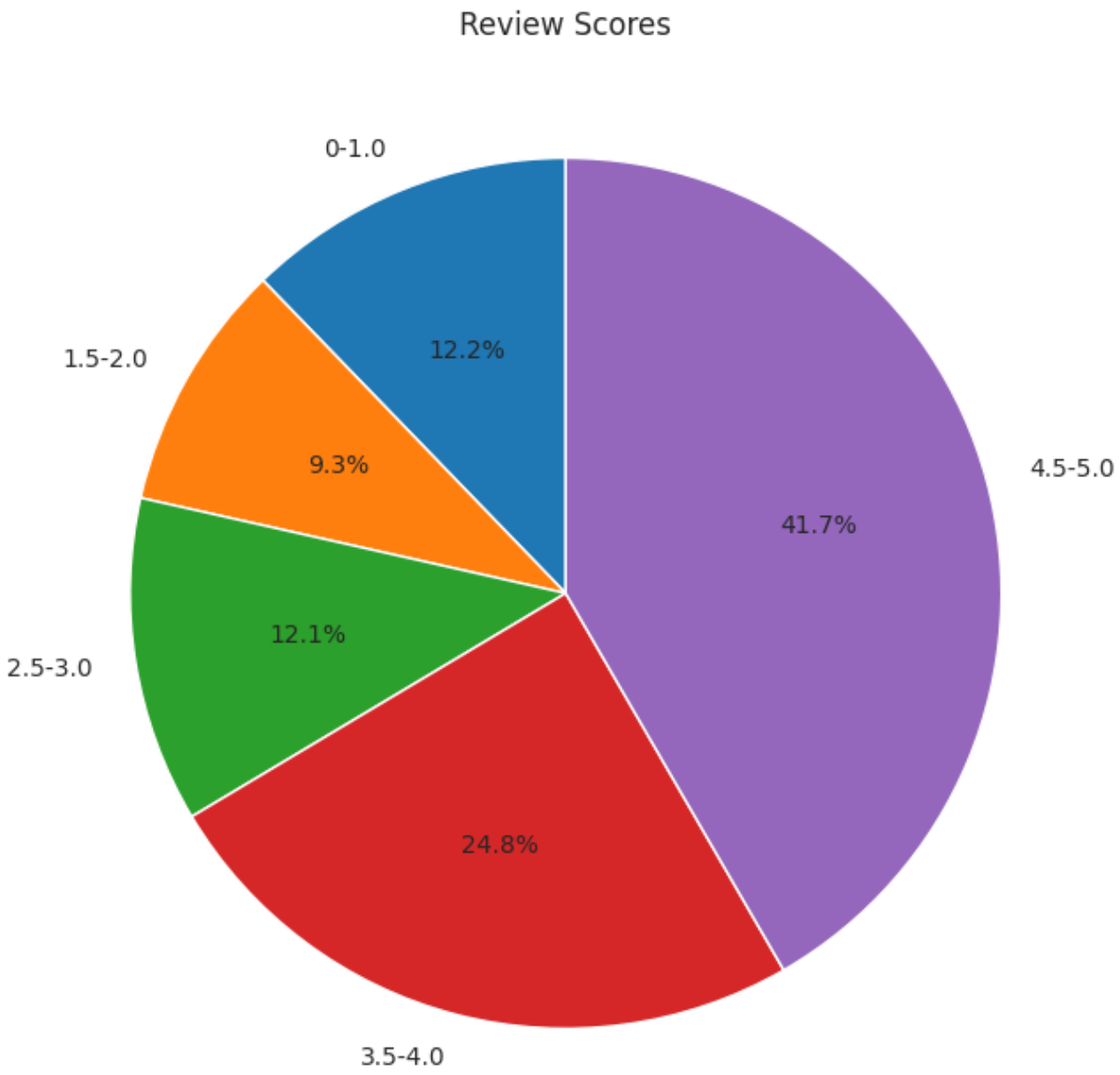


Fig.4., Percentages of Review Scores

The pie chart shows that 41.7% of restaurants have a high rating between 4.5 and 5.0, and the least number of restaurants have a low percentage of 9.3% for the rating of 1.5 to 2.0. This means that most reviews may be deemed as positive. As for tabular visualizations, a table showing grouped the restaurants based on the attribute that is_open was created to retrieve the number of restaurants that are open and closed (Table 1). This was done mainly for the purpose of balancing the final

dataset to avoid class imbalance and to inform what splitting strategy to use. The final number of open businesses is around 95 thousand, while there are about 94 thousand closed businesses.

is_open	c
0	94642
1	95437

Table 1, The distribution/counts of open (1) and closed (0) restaurants

Then, code was written to see how many unique businesses are present in the dataset, since one business can have more than one review. The results showed that there are about 34 thousand unique open businesses and 16 thousand unique closed businesses. In addition, the ten most important terms according to the TF-IDF were visualized as seen in Figure 5.

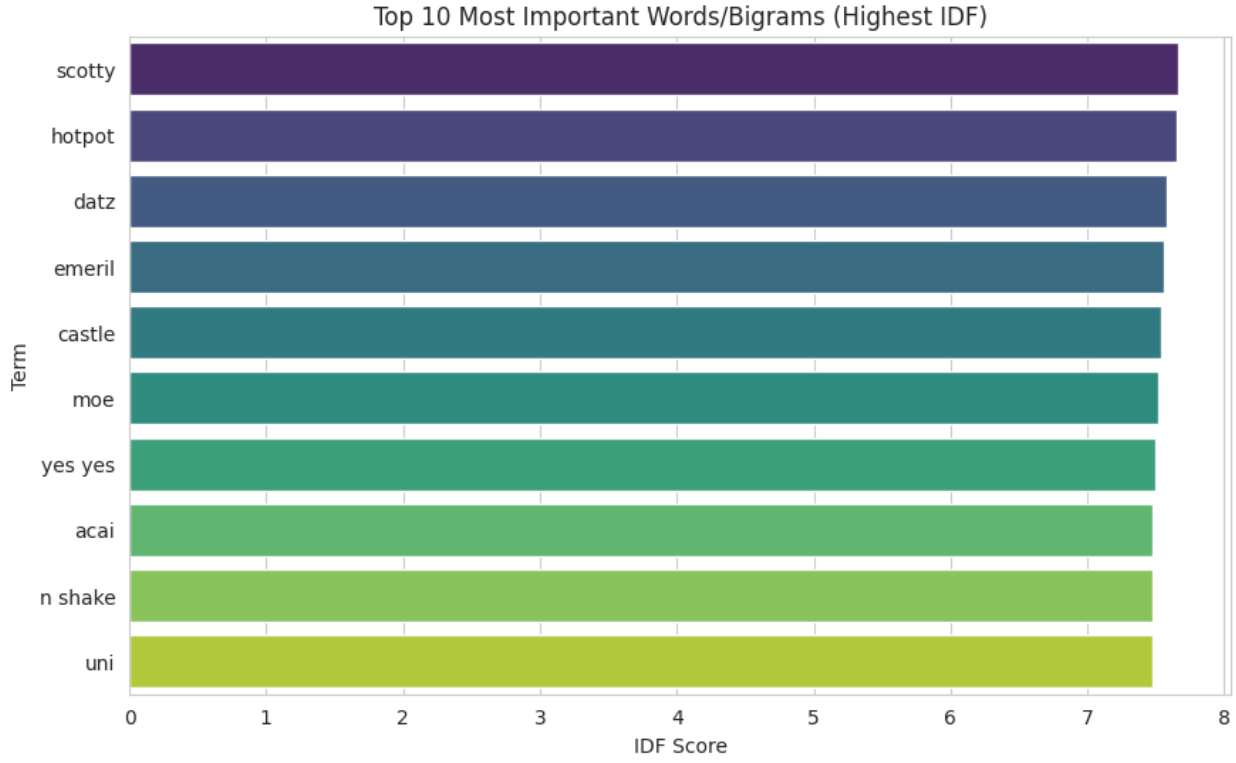


Fig.5., Top 10 Most Important Terms

According to the plot, the most important term is the word 'scotty', with a near 8.0 IDF score. Overall, all the top ten words have shown to have a high IDF score, around 7.5. It can also be

seen that as we move down the list of terms, their score starts to decrease slightly, indicating that the words are of less importance.

e. Feature engineering

For feature engineering, Term Frequency – Inverse Document Frequency (TF-IDF), sentiment analysis, and date encoding was conducted. For TF-IDF, unigrams and bigrams were considered. Unigrams since they are the standard approach, and bigrams since they can capture more context. Then they are combined into one column using VectorAssembler, and IDF is performed. The column is in the form of a sparse vector from which the selected machine models can meaningfully learn. Moreover, since bigrams can capture more context, the word 'not' was removed from the stopwords list. For sentiment analysis, first, a TF-IDF column is built (since sentiment analysis has a different cleaning strategy). Then, three machine learning models are compared to select the best final one. Logistic Regression, Naive Bayes, and Random Forest were implemented. The target column for each was an engineered column named 'stars_review_binary'. It is a binary version of the existing column 'stars_review' which represents the star rating that accompanies each review given by a customer. Since it ranges from 0 to 5, the reviews up to 3 stars are encoded as number 0, and others as number one. Although all models performed relatively well, the best model was Naive Bayes (accuracy of 0.904) by a small margin compared to Logistic Regression (accuracy of 0.895) (Figure 6).

```
Final Results
Logistic Regression: 0.895826738419331
Naive Bayes: 0.9042154690302838
Random Forest: 0.7850585628363406
```

Fig.6., Accuracy of models used for sentiment analysis

For date handling, months and days were encoded using sine and cosine transformations so the models can learn the cyclical patterns, while years are extracted to discover linear patterns. This was done for both the date the review was left and for the last date checkin which is tied to the restaurant itself. Lastly, after the 80/20 stratified split (to ensure the open to closed restaurant ratio is maintained), according to the 'business_id' to avoid data leakage, the top 10 most important terms were extracted from the TF-IDF.

f. Build models

Before creating the models used in the project, a literature review was conducted to select models best suited for the project. The resulting models were identified:

- Random Forest
- Support Vector Machines (SVM)
- Logistic Regression

The three models are initialized with mostly default parameters, with a few intentional adjustments. The column names are renamed across all models to avoid conflicts. In LinearSVC (SVM), regParam is set to 0.1 instead of the default 0.0, adding a small amount of L2 regularization to penalize large weights and reduce the risk of overfitting. In Random Forest, the number of trees was set to 20 while the maximum tree depth is set to 10. This was a decision made due to the dataset size as the default configuration was likely to be too conservative. Dictionaries keeping track of the trained models and their corresponding predictions are implemented so that the best model can be retrieved and saved.

g. Evaluate and present results

For evaluation, a custom function was written so that all required and relevant metrics are more easily calculated. By using both MulticlassClassificationEvaluator and BinaryClassificationEvaluator, the function extracts a multi-dimensional view of model efficacy, specifically capturing the Area Under the ROC (AUC-ROC) to measure discriminative ability, alongside accuracy, F1-score, weighted precision, and weighted recall. These were chosen as the dataset was balanced (no need for balanced accuracy, for example) and classification, rather than regression, was performed. Table 2 shows the overview of the metrics of each model.

Model	AUC-ROC	Accuracy	F1-score	Precision	Recall
SVM	0.9544	0.8872	0.8872	0.8873	0.8872
RF	0.8302	0.7430	0.7421	0.7460	0.7430
LogReg	0.9566	0.8991	0.8991	0.8991	0.8991

Table 2, Evaluation metrics of each model

After the final model was selected, Logistic Regression, an analysis was made about which restaurants were incorrectly identified to be open but actually closed. An example review of such a restaurant: “Good food, quickly. Service is always top notch, and it's exactly what you want, when you want it!” Then, two visualizations were made to check the model performance. Figure 7 shows the model accuracy by review length, while Figure 8 shows the total correct against the incorrect predictions.

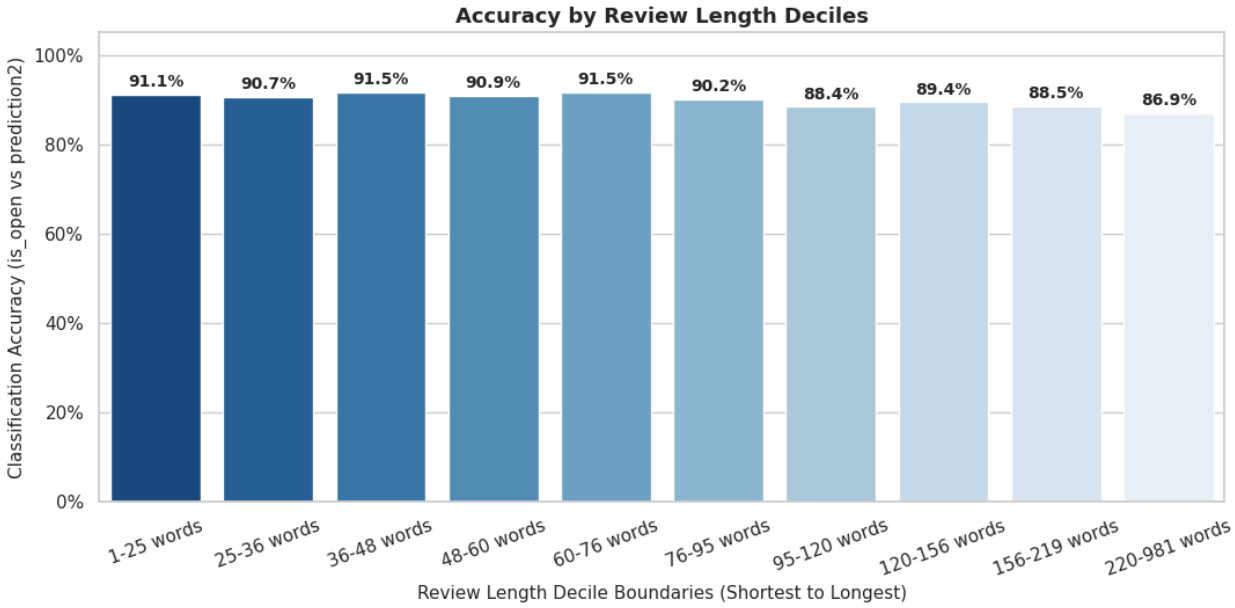


Fig.7., Model accuracy by review length

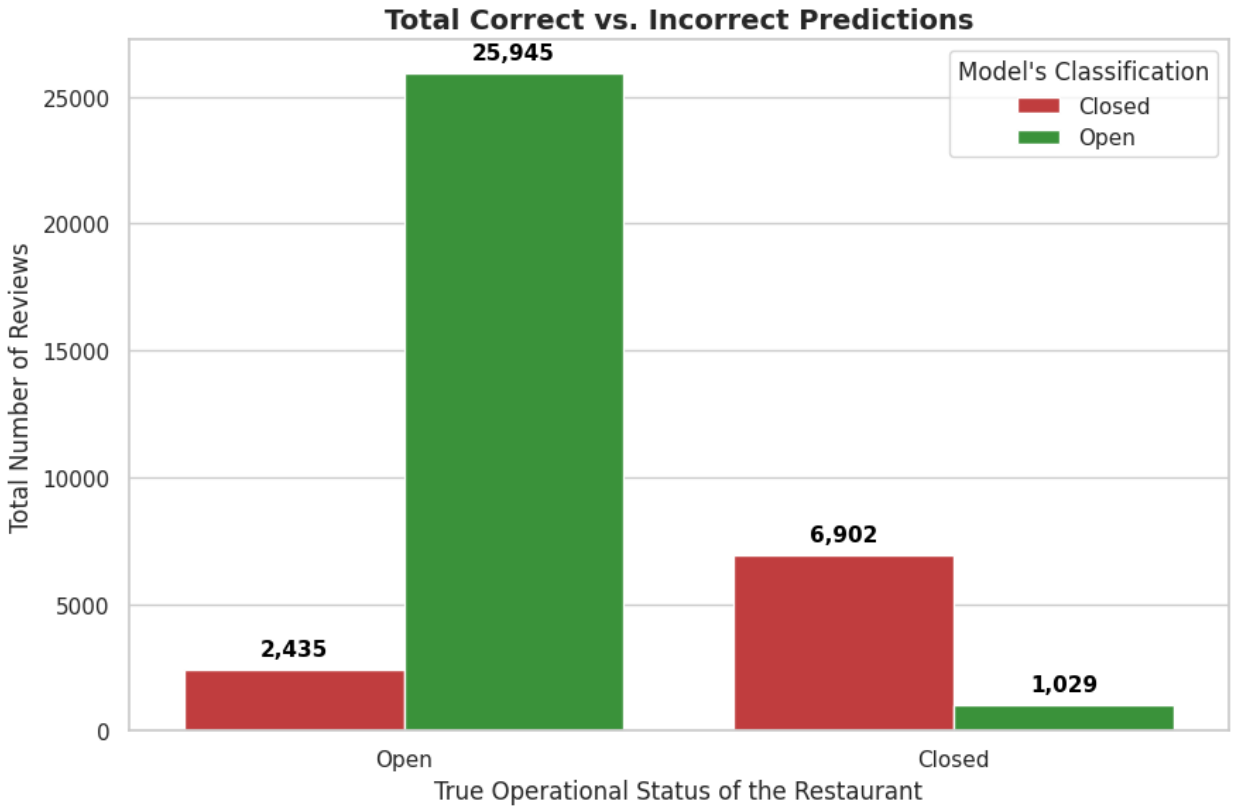


Fig.8., Total correct vs. Incorrect predictions

Technology used

The project was done and executed in Google Colab, using Google Disk to store the used dataset files available for easy and quick access. The distributive computing environment is powered by PySpark (Apache Spark), which was initialized through a local SparkSession. For the processing of big data, libraries such as PySpark SQL were imported and used for loading, filtering, joining, and transforming the JSON datasets via Spark DataFrame and SQL queries for joining the datasets together. A couple of additional libraries were imported for creating plots and charts for the Exploratory Data Analysis, such as matplotlib and seaborn. For Natural Language Processing, various modules were imported to enable the process of tonkenization, removing stopwords, creating n-grams, as well providing the ability to perform the actual TF-IDF and sentiment analysis.

Discussion and Future Work

The importance of the work can be seen in the usage by restaurant owners and managers for more informed customer feedback. As the model predicts from customer reviews whether the restaurant will close, and not from location, type of restaurant, etc., it provides useful insight about the business's success. Additionally, as the approach uses simpler techniques such as TF-IDF and Sentiment Analysis, rather than deep learning, the model can be easily implemented. This shows that even less sophisticated techniques can produce high results, such as the Logistic Regression achieving an AUC-ROC of 0.9566 and an accuracy of 0.8991. The Support Vector Machine follows closely behind, whereas the Random Forest model shows a performance drop. Since the linear models significantly outperform the tree-based model, the underlying dataset features are likely high-dimensional and linearly separable. Furthermore, the identical values shared between accuracy, F1-score, and recall for each model indicate a highly balanced dataset, while the high AUC-ROC scores confirm that all three models have excellent class-separation. This aligns with the assumption that reviews can be a valuable indicator of restaurant success. Moreover, the good model performance for sentiment analysis (all above 0.78 accuracy) indicates that reviews often have a strong positive or negative sentiment. Logically, most people are not likely to leave a neutral review. Additionally, when analyzing which restaurants were predicted to stay open but actually closed, the reviews were positive, showing that the model failed for those restaurants that closed for reasons not related to their service or customer satisfaction (e.g. moving to another location, natural disasters, economic crises, etc.), and that those restaurants were in the minority (given the high model performance). The final model evaluation also demonstrates stability across different levels of textual input (7). The pipeline maintains a high classification accuracy between 86.9% and 91.5%, regardless of the word count of the customer feedback. In Figure 8, the model shows good performance in predicting open businesses (25,945 true positives), while also correctly identifying closures (6,902 true negatives). The low counts of false positives (1,029) and false negatives (2,435) confirm the pipeline's reliability. However, there are several areas that can be improved in the future. For instance, sentiment analysis may be more informative if a ‘trend’ feature were created, which showed the sentiment change over time. Additionally, running the code on distributed hardware for scalability, so more comprehensive tuning and more complex feature engineering can be done. More models could also be evaluated such as XGBoost, which was identified through a literature review.

Conclusion

The project has successfully used machine learning and NLP to predict a potential restaurant failure from online Yelp reviews. Logistic Regression showed to be best performed, achieving an 89% accuracy, outperforming SVM and Random Forest. These results confirm that a customer's sentiment in restaurant reviews is an important and strong indicator of a restaurant's success or failure.

AI Use Disclosure

The use of AI was limited. It was primarily used to create visualizations and to help understand what the functions should deliver and in what way. Additionally, AI was used to streamline the debugging process, but the changes were thoroughly understood and applied manually. The code was optimized by AI to create the final metrics table. The report was written manually.

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Appendix A

Code (Github link): https://github.com/normala127/NLP_Yelp_Review_Project

Original dataset source: https://business.yelp.com/data/resources/open-dataset/

Final dataset (100 random rows; CSV format) (Drive link): https://drive.google.com/file/d/1qqKQlTyt2EBh1hn379iZvMlcXjhzM6C3/view?usp=sharing